Scaleform GFx

Scaleform LITE Customization

This document describes how to customize Scaleform to a lightweight configuration, Scaleform LITE.

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Autodesk® Scaleform® 4

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Table of Contents

1 Scaleform 4.0 Customization Defines 1

1.1 AS3 Class Registration 1

2 Building and Using the “Lite” Version 2

2.1 General Options 2

2.2 Rendering Related Options 6

2.3 Logging Related Options 7

2.4 Font Related Options 8

2.5 ActionScript Related Options 9

1 Scaleform 4.0 Customization Defines

Autodesk® Scaleform® is highly customizable and allows users to exclude many modules and functionalities in order to decrease the code size. This is especially important for game consoles and the Wii™ in particular, since it has a strict limitation on executable code size. This document details how to customize and use the lightweight version of Scaleform, known as "Scaleform LITE" and the various options defined in this version. Using the lightweight version of Scaleform, customers can build their own customized Scaleform libraries and achieve significant code size reduction in their size-critical console applications.

All the possible options are represented as “defines” in the header file Include/GFxConfig.h. Scaleform provides two variants of the Scaleform library – regular (or full) and “lite”. The “lite” version doesn’t contain some features that the full version contains in order to decrease the code size.

Since modifying these defines requires source code changes and recompilation, this type of customization is available for source code customers only.

1.1 AS3 Class Registration

Scaleform 4.0 can also be customized by specifying exactly which AS3 classes are needed by the application. This is yet another way of using a lighter-weight version of the library. This is done by modifying the file “GFx\AS3\AS3_Global.h in the Src folder and commenting out the un-wanted classes and defines. The file gives developers some flexibility by providing pre-existing defines such as SF_AS3_VERSION_MULTITOUCH, SF_AS3_VERSION_XML, SF_AS3_VERSION_SOUND, SF_AS3_VERSION_NET, SF_AS3_VERSION_VIDEO, and SF_AS3_VERSION_IME.

Note: The inclusion of this file is mandatory for any application which uses Scaleform and AS3, since it defines the AS3 class registration table. This file should only be included once. To include the header file directly in the application class, use the following:

```
#include "GFx/AS3/AS3_Global.h"
```

Modification of the AS3 class registration table in AS3_Global.h does not require access to Scaleform 4.0 source code and can be utilized by binary customers as well.

2 Building and Using the “Lite” Version

To build the “Lite” version of Scaleform library, it is necessary to define a symbol ‘GFX_BUILD_LITE’ and rebuild Scaleform and all corresponding libraries and source files.

If the “Lite” version of Scaleform library is used, it is necessary to define the symbol ‘GFX_BUILD_LITE’ for the application project as well. This will make sure the same set of options is consistent between the application and all related modules during compilation. Otherwise, linker or compiler errors might show up.

Note that if using GFxPlayerTiny as the Scaleform player in Lite configurations, customers have to convert *Bin\Samples\Window.swf* using GFxExport into GFx format for use.

```
GFxExport_Release Window.swf -i DDS -d0 –gradients
```

As a result of this conversion, these files would be created: Window.gfx, Window_G0.dds, Window_G1.dds and Window_G2.dds. Use the *Window.gfx* instead of the original *Window.swf* as an input Flash file for GFxPlayerTiny for Lite configuration.

Below, each section contains a list of the options, grouped by category, along with a brief description and whether the option is defined in the regular and lite libraries.

2.1 General Options

SF_ENABLE_THREADS
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Un-define this macro to disable thread support. This option will also disable progressive loading and video playback.

SF_ENABLE_STATS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Disables statistics tracking if un-defined; this is useful for the final build.

SF_NO_DOUBLE
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: This macro needs to be defined if it is necessary to avoid the use of the “double” type – double precision floating point type. Some platforms support only single precision – “float”. In that

case, all references to “double” will be replaced by “float”. However, this substitution might cause some unexpected results in ActionScript logic, especially when arithmetic operations are followed by comparisons for equality with hardcoded values.
NOTE: By default, PS2 and PSP define this if not manually defined here.

SF_ENABLE_LIBJPEG
Regular Scaleform: Defined
“Lite” Scaleform: Not defined
Description: Un-define this macro to disable use of LIBJPEG and make JPEGUtil a no-op stub. If disabled, SWF JPEG image loading will stop functioning.

SF_CPP_LIBJPEG
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Define this macro if the whole JPEGLIB is compiled as C++ code. By default, libjpeg is a pure C library and public names are not mangled. Though, it might be necessary to mangle jpeglib's names in order to resolve names clashing issues (for example, with XBox360's xmedia.lib).

SF_ENABLE_ZLIB
Regular Scaleform: Defined
“Lite” Scaleform: Not defined
Description: Un-define this macro to disable use of ZLIB and comment out GFx::ZLibFile class. If ZLIB is disabled, compressed SWF and GFX files will no longer load, as well as the lossless images embedded into SWF files. Use this option only if GFxExport is used to extract all images from the SWF file.

SF_ENABLE_LIBPNG
Regular Scaleform: Defined
“Lite” Scaleform: Not defined
Description: Enables use of LIBPNG. If disabled, SWF PNG image loading will not function.

SF_NO_WCTYPE
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Define this macro to eliminate custom wctype tables for functions like G_iswspace, SFtowlower, SFtowupper and so on. If this macro is defined, Scaleform will use system Unicode functions (which are incredibly slow on Microsoft Windows and work incorrectly on consoles).

GFX_AS_ENABLE_GC
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined

Description: Enable garbage collection for ActionScript. Note, if garbage collection is disabled then Scaleform may produce memory leaks in the case of circular references.
Here is an example of code that will produce a leak in the case where garbage collection is disabled, unless one of the object references is explicitly disconnected:

```
var o1 = new Object;
var o2 = new Object;
o1.a = o2;
o2.a = o1;
```

GFX_ASSERT_ON_GRADIENT_BITMAP_GEN
Regular Scaleform: Not defined
"Lite" Scaleform: Not defined
Description: Define this macro to throw an assertion if any gradient texture is generated during the runtime. It helps detect potential slow-downs in these type of operations on low-end platforms. GFxExport with the option "–gradients" should be used to avoid this assertion, if the macro is defined.

GFX_ASSERT_ON_RENDERER_RESAMPLING
Regular Scaleform: Not defined
"Lite" Scaleform: Not defined
Description: Define this macro to throw an assertion if any re-sampling occurred in the renderer during the runtime. It helps detect potential slow-downs on these types of operations on low-end platforms. The renderer may resample if non-power-of-2 textures are used on hardware that doesn't support it. Use GFxExport to avoid this assertion when the macro is defined.

GFX_ASSERT_ON_RENDERER_MIPMAP_GEN
Regular Scaleform: Not defined
"Lite"Scaleform: Not defined
Description: Define this macro to throw an assertion if any mipmap level generation occurred in the renderer during the runtime. It helps detect potential slow-downs in these type of operations on low-end platforms. Use GFxExport with the option "–i dds" to pre-generate mipmaps and to avoid this assertion when the macro is defined.

GFX_ENABLE_GRADIENT_GEN
Regular Scaleform: Not defined
"Lite" Scaleform: Not defined
Description: Define this macro to exclude gradient generation. If this option is defined, then gradient generation code is excluded from the build. GFxExport with the option "-gradients" should be used to avoid gradient rendering problems. The option **GFX_ASSERT_ON_GRADIENT_BITMAP_GEN** can be enabled to detect any attempts to generate gradients during run-time.

GFX_ENABLE_SOUND
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Define this macro to enable sound support (in the Scaleform core and ActionScript).

GFX_ENABLE_VIDEO
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Define this macro to enable video support (in the Scaleform core and ActionScript). Note, video support is automatically disabled if the macro SF_ENABLE_THREADS is defined, since video can be used only with multithreading support.

GFX_ENABLE_CSS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable core and ActionScript CSS (style sheets) support.

GFX_ENABLE_XML
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable core and ActionScript XML support.

SF_NO_IME_SUPORT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Disable core and ActionScript IME support. If IME is disabled, then typing in Asian languages (Japanese, Korean, and Chinese) will be impossible. If Asian language typing is not required (the IME library is not used) or **GFX_ENABLE_TEXT_INPUT / GFX_ENABLE_KEYBOARD** options are enabled, then it is safe to disable IME support.

GFX_ENABLE_BUILTIN_KOREAN_IME
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable the built-in core Korean IME logic. This option affects only the Windows Scaleform platform. Built-in Korean IME allows users to type in the Korean language without using a separate IME library. It is safe to disable this feature if Korean support is not required or if **GFX_ENABLE_TEXT_INPUT / GFX_ENABLE_KEYBOARD** options are enabled.

GFX_ENABLE_TEXTFIELD_EXTENSIONS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined

Description: Enable TextField ActionScript extension functions. If this option is disabled then the standard Scaleform Player's HUD will not work.

GFX_ENABLE_TEXT_INPUT
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined; defined for consoles
Description: Enable text editing. When disabled, text selection will be disabled as well (since it is a part of text editing). This macro might be useful on consoles.

GFX_ENABLE_DRAWTEXT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable the DrawText API.

GFX_ENABLE_KEYBOARD
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Enable keyboard support. When disabled, no Key AS class will be provided, HandleEvent with
GFx::KeyEvent will not be supported; PAD keys on consoles will not work as well.

GFX_ENABLE_MOUSE
Regular Scaleform: Not defined.
“Lite” Scaleform: Defined for consoles other than Wii.
Description: Enable mouse support completely. When disabled, this option also disables the Mouse AS class (see **GFX_AS2_ENABLE_MOUSE**).

2.2 Rendering Related Options

SF_RENDER_ENABLE_EDGEAA
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: If this macro is defined, Scaleform will include EdgeAA (anti-aliasing) support.

SF_RENDER_ENABLE_MORPHING
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Enable morphing (shape tween) support.

SF_RENDER_ENABLE_GLYPH_CACHE
Regular Scaleform: Not defined
"Lite" Scaleform: Not defined
Description: Enable this macro to include the dynamic glyph cache.

2.3 Logging Related Options

GFX_VERBOSE_PARSE
Regular Scaleform: Not defined
"Lite" Scaleform: Defined
Description: Define this macro to enable support for verbose parsing of input files. If this option is set, none of the verbose parse options are available, and the GFx::Loader::SetVerboseParse call will have no effect. Game production release builds should probably define this option.

GFX_ VERBOSE_PARSE_ACTION
Regular Scaleform: Defined if GFX_VERBOSE_PARSE is defined
"Lite" Scaleform: Defined
Description: Define this macro to enable support for verbose parsing of actions (disables support for logging ActionScript disassembly during loading).

GFX_VERBOSE_PARSE_SHAPE
Regular Scaleform: Not defined; defined if GFX_VERBOSE_PARSE is defined
"Lite" Scaleform: Defined
Description: Define this macro to enable support for verbose parsing of shape character structures.

GFX_VERBOSE_PARSE_MORPHSHAPE
Regular Scaleform: Not defined; defined if GFX_VERBOSE_PARSE is defined
"Lite" Scaleform: Defined
Description: Define this macro to enable support for verbose parsing of morph shape character structures.

GFX_AS2_VERBOSE/GFX_AS3_VERBOSE
Regular Scaleform: Not defined in Debug and Debug Opt configuration; defined in Release configuration
"Lite" Scaleform: Defined
Description: Define this macro to enable support for verbose logging of executed ActionScript opcodes. If this macro is undefined, GFx::Movie::SetVerboseAction will have no effect. This option is defined by default in the Release configuration.

GFX_AS2_VERBOSE_ERRORS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Define this macro to enable support for verbose logging of ActionScript run-time errors. If this macro is undefined, GFx::Movie::SetVerboseActionErrors will have no effect. Game production release builds should probably undefine this option.

2.4 Font Related Options

GFX_ENABLE_GLYPH_PACKER
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Define this macro to exclude the Font Glyph Packer. The Font Glyph Packer is used only if the static font cache is used (when `"Loader.GetFontCacheManager()->EnableDynamicCache(false);"` is used). If the dynamic font cache is in use or if GFxExport with option “-fonts” was used to produce the .gfx file with pre-rendered font textures, then the Font Glyph Packer may be excluded safely, in order to save some code size.

GFX_ENABLE_FONT_COMPACTOR
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable the font compactor (compaction during the run-time). When disabled, fonts compacted by the GFxExport (with option “ -fc”) will work.

GFX_ENABLE_COMPACTED_FONTS
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Enable usage of compacted fonts (fonts, compacted by GFxExport (option “-fc”)).

GFX_ASSERT_ON_FONT_BITMAP_GEN
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Define this macro to throw an assertion if any font texture is generated during the runtime. This option is useful to detect any run-time font texture generation for low-end platforms. GFxExport with the option “–fonts” should be used to avoid this assertion, if the macro is defined.

2.5 ActionScript Related Options

GFX_AS2_ENABLE_FILTERS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable filter classes support, such as ‘flash.filters.DropShadowFilter’, ‘flash.filters.BlurFilter’, ‘flash.filters.BitmapFilter’, ‘flash.filters.GlowFilter’. Note, currently these classes are supported only for TextFields.

GFX_AS2_ENABLE_DATE
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'Date' ActionScript class support.

GFX_AS2_ENABLE_POINT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'flash.geom.Point' ActionScript class support. If the ‘Point’ class is used by another class (for example by ‘flash.geom.Rectangle’) then it will replaced by a regular Object with the members “x” and “y” set.

GFX_AS2_ENABLE_RECTANGLE
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'flash.geom.Rectangle' ActionScript class support. If the ‘Rectangle’ class is used by another class (for example by ‘Stage’) then it will be replaced by a regular Object with the members “x”, “y”, “width”, “height” set.

GFX_AS2_ENABLE_TRANSFORM
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'flash.geom.Transform' ActionScript class support.

GFX_AS2_ENABLE_COLORTRANSFORM
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'flash.geom.ColorTransform' ActionScript class support.

GFX_AS2_ENABLE_MATRIX
Regular Scaleform: Not defined
“Lite” Scaleform: Defined

Description: Enable 'flash.geom.Matrix' ActionScript class support.

GFX_AS2_ENABLE_TEXTSNAPSHOT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'TextSnapshot' class support.

GFX_AS2_ENABLE_SHAREDOBJECT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'SharedObject' class support.

GFX_AS2_ENABLE_MOVIECLIPLOADER
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'MovieClipLoader' ActionScript class support.

GFX_AS2_ENABLE_LOADVARS
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'LoadVars' ActionScript class support.

GFX_AS2_ENABLE_BITMAPDATA
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'flash.display.BitmapData' ActionScript class support. Note, if the ‘BitmapData’ class is disabled then textfield will not support <IMG> HTML tags and image substitutions; ‘MovieClip.attachBitmap’ won’t work as well.

GFX_AS2_ENABLE_CAPABILITES
Regular Scaleform: Not defined
“Lite”Scaleform: Defined
Description: Enable 'System.capabilites' ActionScript class support.

GFX_AS2_ENABLE_COLOR
Regular Scaleform: Not defined.
“Lite” Scaleform: Defined
Description: Enable 'Color' ActionScript class support.

GFX_AS2_ENABLE_TEXTFORMAT
Regular Scaleform: Not defined
“Lite” Scaleform: Defined

Description: Enable 'TextFormat' ActionScript class support.

GFX_AS2_ENABLE_SELECTION
Regular Scaleform: Not defined
“Lite” Scaleform: Defined
Description: Enable 'Selection' ActionScript class support.

GFX_AS2_ENABLE_STAGE
Regular Scaleform: Not defined
“Lite” Scaleform: Not defined
Description: Enable 'Stage' ActionScript class support. If disabled, Stage.height and Stage.width will not be supported as well.

GFX_AS2_ENABLE_MOUSE
Regular Scaleform: Not defined
“Lite” Scaleform: Defined only for consoles other than Wii
Description: Enable 'Mouse' ActionScript class support.